30. Cultural studies of collaborative science and technology

Abstract

A cultural approach to interdisciplinary and transdisciplinary research focuses on how these modes of research can more effectively address scientific and societal challenges. Worldwide, multidimensional challenges, such as the pandemic, climate change or migration crises, urgently call for more collaborative research that can help in transforming social reality. However, many factors still act as obstacles to high-impact research, resulting in deficiencies and disconnects between practice and policy. As a result, inter- and transdisciplinarity are trapped in a pervasive contradiction between promotional rhetoric, on the one hand, and relatively inflexible institutional and funding reward systems, on the other hand. This contradiction needs to be fully elaborated on to understand the political, societal and economic values underlying inter- and transdisciplinarity. To fill this gap, the research programme on cultural studies of collaborative science and technology, which focuses on inter- and transdisciplinarity, studies the interconnections among i) those who study inter- and transdisciplinary cultures, ii) those who engage in inter- and transdisciplinary research and iii) those who promote and fund inter- and transdisciplinary research.

Keywords

Cultural approach; Problematisation; Knowledge regimes; Critical perspective; Science policy; Collaborative science

Mapping the challenges

Worldwide, multidimensional challenges, such as the pandemic, climate change or migration crises, urgently call for more collaborative research that can help in transforming social reality. Collaboration entails the following:

> Boundary weaving through dialogue in a communication boundary space that develops a cooperation and communication culture. For collective identity to emerge in a group, collective communication competence is crucial, facilitated by socio-cognitive platforms and collaborative interdisciplinary reasoning for integration, communicative rationality, and communication action. (Klein, 2021, p. xvi)

However, many factors still act as obstacles to high-impact research, resulting in deficiencies and disconnects between practice and policy. As a result, inter- and transdisciplinarity, as specific formats of collaboration, are trapped in a pervasive contradiction between promotional rhetoric, on the one hand, and relatively inflexible institutional and funding reward systems, on the other. This gap needs to be fully elaborated on to understand the political, societal and economic values that underlie inter- and transdisciplinarity. If inter- and transdisciplinarity remain elusive, is it because we are still incapable of methodologically grasping their heterogeneity and contextual differences? There is an urgent need for an in-depth understanding of what it means to do inter- and transdisciplinarity, with their epistemic, cultural, social and political implications for the science–policy interface.

The contradiction between promotional rhetoric and relatively inflexible reward systems has three interrelated dimensions conceptualised as inter- and transdisciplinary: i) cultures; ii) practices; and iii) policies. In relation to *cultures*, efforts to categorise the processes and outcomes of collaborative research depend on an array of interrelations and negotiations between academic disciplines, as well as between those disciplines and bodies of knowledge contained in programmatic statements, policy interventions, institutional forms, theoretical statements, instruments, materials and research practices (Barry & Born, 2013).

It is necessary to find connections among existing conceptualisations of inter- and transdisciplinarity because these represent different research *cultures* (in Bauman's [1999] sense of "culture as praxis" and Hess's [1992] conceptualisation of a "scientific culture"). In addition, there have been increasing calls for new ways of producing knowledge through inter- and transdisciplinary research. Historical accounts show that inter- and transdisciplinarity have made profound contributions to both the scientific and policy realms. These different accounts show

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that a lot of reiteration and relearning takes place (Bammer et al., 2020).

At the same time, those *practising* inter- and transdisciplinarity face multidimensional challenges, including a lack of academic legitimation, contested understandings of their research, dispersed scientific communities and weak institutional career support. To respond to these challenges, previous studies have focused on two main aspects: i) how to perform inter- and transdisciplinary research and what factors hinder or help their development; and ii) the best methods and tools for developing both inter- and transdisciplinary research. A topic arising from this literature is the lack of an organised body of knowledge that would help scholars and funders better understand the urgent problems faced by inter- and transdisciplinary researchers and practitioners (Bammer et al., 2020).

The third dimension refers to the mismatch between *policy discourses* promoting inter- and transdisciplinarity, on the one hand, and, on the other, the relatively inflexible government systems for funding and evaluating research. This misalignment between discourses materialises in the proliferation of a programmatic and instrumental perspective of inter- and transdisciplinarity in the policy literature. The latter is accompanied by modes of governance based on monodisciplinary systems of research funding and evaluation. As a result, scientific and policy discourses on inter- and transdisciplinarity have pursued different objectives. The policy discourse usually ignores the fact that inter- and transdisciplinary research employ a wide range of institutional arrangements, procedures and methods, in which different ways of conducting of inter- and transdisciplinarity lead to different types of impacts (Vienni-Baptista et al., 2022).

The evidence cited above has led scholars in different fields to appreciate that more research on inter- and transdisciplinarity is necessary. Several scholars have noted that empirical examinations of inter- and transdisciplinarity often take the form of individual case studies, thus limiting the comparability and generalisation of results across cases (Vienni-Baptista, 2022). Because of this tendency, the current understanding of inter- and transdisciplinary research practices is still incomplete, with little potential for comparison and a disconnection between theory and practice. Other authors also warn about the risk of promoting "fake collaborations" if there is no robust foundation on which to sustain funding policies. Funders have perceived the same problem, noting that studies on inter- and transdisciplinarity are demanded by the policy sector. Nevertheless, recommendations in the academic and policy realms repeatedly point to a lack of joint learning, resulting from the weak links between academic and policy literature (Vienni-Baptista et al., 2022).

It can be concluded that the contradiction between promotional rhetoric and funding reward systems on inter- and transdisciplinarity leads to fragmented responses to demands from policymakers, funders and practitioners. More high-quality, inter- and transdisciplinary collaborative research is critically needed, together with more in-depth knowledge about what it means to carry out inter- or transdisciplinary research, how and where to perform such research, why and how to professionalise inter- or transdisciplinary research and how to support them.

A research programme on cultural studies of science and technology

As an alternative pathway to the challenges elaborated in the previous section, I propose a research programme on cultural studies of collaborative science and technology to study the intersections among: i) those who study inter- and transdisciplinary cultures, ii) those who engage in inter- and transdisciplinary research practices, and iii) those who promote and fund inter- and transdisciplinary research and policies. This approach seeks to improve the responsiveness of inter- and transdisciplinarity to demands to address societal challenges. This stems from the subfield of the anthropology of science within science and technology studies (STS). The anthropology of science applies a cultural perspective that attempts "to understand the world from the 'native's point of view'" (Geertz, 1973, p. 5) and "for which the background framework is anthropology's tradition of the comparative study of cultures" (Hess, 1992, p. 2). An anthropological study of inter- and transdisciplinarity places these − as well as science − within national and international research cultures, showing that these are not naturally given but rather temporal and spatial symbolically and politically construed.

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The cultural approach strongly builds on the concept of *intersections*, which are forms of cultural meaning that emphasise how things are partially connected and, hence, kept apart by parameters of belonging, difference and identity (Edwards, 2000). A lack of a deep understanding of these intersections leads to pockets of undone science, in which different problem spaces are unequally investigated (Hess, 2016). This concept of undone science draws attention to a kind of non-knowledge or ignorance that is systematically produced through the unequal distribution of power in society (Hess, 2016), being a questioning and decentring approach.

The research programme on cultural studies of collaborative science and technology implies three study areas that contribute to the development and further investigation of inter- and transdisciplinarity that I explain in the following sections (Vienni-Baptista, 2022).

Providing a critical perspective on collaborative science and technology

Providing a critical perspective on collaborative science and technology entails disentangling inter- and transdisciplinary culture(s). The starting point of this line of research is the comingling of anthropology with other disciplines within the cultural studies of science and technology. This multimethod cultural critique includes comparative and textual methods, taking as the unit of analysis research conducted within inter- and transdisciplinary communities (Hess, 1992). This critical approach seeks to elucidate the system of values that make the discourse and actions of inter- and transdisciplinary researchers meaningful, as well as the values that pertain to the larger social, political and institutional contexts enacted by policymakers and funders. For example, to unfold the complementarities between the discourses on transdisciplinarity and transformations, we critically examined two case studies using auto-ethnography (see details in Alvial Palavicino et al., 2023).

This first study area focuses on the cultural dimensions of inter- and transdisciplinarity, here guided by the question, "How are inter- and transdisciplinarity theorised?" In a related vein, discourses on inter- and transdisciplinarity reflect different aspects of the science–society relationship, including the *epistemological*, the *conceptual*, the *cultural*, the *institutional*, the *collaborative* and the *methodological*. Studying inter- and transdisciplinary practices and policies from an anthropological perspective implies a thick description (Geertz, 1973) of the intersections of such discourses.

Problematising concepts and methods to study collaborative research, its output and its impact

The second study area focuses on how to investigate the implications and impacts of inter- and transdisciplinarity as a means to epistemological and methodological change and transformation. By problematising the concepts and methods used to study collaborative research and its output and impacts, this research programme contributes to bridging scientific and policy discourses. As understood here, problematisation is defined as a mode of inquiry because it constitutes a questioning and decentring approach (Foucault, 1984). The search for the underlying patterns and meanings that give coherence to people's practices is central (Vienni-Baptista et al., 2022).

Accounts of interdisciplinary or transdisciplinary research usually pay insufficient analytical attention to the sustained push from administrators, policymakers and funding agencies to build new collaborations across disciplines. As a result, the current understanding of inter- and transdisciplinary research practices is still sketchy, with little comparative potential (Vienni-Baptista et al., 2022). Relevant to the study of collaborative research *practices* is the so-called *begleitforschung* (usually translated as "accompanying research") (Defila & Di Giulio, 2018). Accompanying research takes place in parallel with a specific project, acting as a point of reference for developing research and promoting knowledge about the collaboration while it is ongoing. Collectively, these provide a key foundation because they unequivocally confirm the limitations of some research methods when investigating inter- and transdisciplinarity.

Understanding inter- and transdisciplinarity as knowledge regimes to disentangle the intersections of cultures, practices and policies

The mismatch between academic and policy discourses entails a set of implications for the

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development of inter- and transdisciplinarity in science and policy: i) researchers try to frame their problem and research questions according to the mandates of the policy discourse; ii) as a result, inter- and transdisciplinarity do not achieve their full potential in the scientific realm because they are poorly understood and badly supported by the policy sector; iii) the role that policymakers and funders play reinscribes and reinforces a weak support for inter- and transdisciplinarity; and iv) the way that inter- and transdisciplinary research are valued affects their status within institutions (Vienni-Baptista et al., 2022).

The third study area addresses the disconnection between practices and policies by considering inter- and transdisciplinarity as knowledge regimes (here following Felt et al., 2016) and by explaining the intersections of cultures, practices and policies. Inter- and transdisciplinarity are heterogeneous assemblages of three components: i) ideologies and myths guiding inter- and transdisciplinary knowledge production; ii) institutions and their institutional logics (i.e., shared beliefs and practices and broader imaginaries and values embedded in knowledge generation); and iii) researchers, research partners and societal actors who govern research (Felt et al., 2016).

By considering inter- and transdisciplinarity as knowledge regimes, there is a two-way development of steadily stronger interrelationships and mutual influences between the scientific and policy domains that affect discourses on collaborative research. These dialectical dynamics in scientific and policy practices show divergent constructions of the time frames and spaces of knowledge production. For example, we designed a heuristic tool to disentangle the different understandings that inter- and transdisciplinarity have in academia and in policy (Vienni-Baptista & Pohl, 2023). The tool is a practical means to connect these discourses and find intersections in practice and policy. Thus, "intersections" are deemed relevant by paying analytical attention to practices of inter- and transdisciplinary research and how to bridge them with administrators, policymakers and funding agencies to build new collaborations to approach societal challenges.

Conclusion

A cultural approach to inter- and transdisciplinarity provides the foundations for moving the field forward in three main areas (Vienni-Baptista, 2022):

1. By refining and expanding the theories and methods used to investigate inter- and transdisciplinary knowledge *cultures*, we can advance the theoretical and methodological foundations for investigating inter- and transdisciplinarity as collaborative practices in research and teaching to leverage their impacts. In this area, future research may focus on designing and testing new tools and frameworks to better investigate inter- and transdisciplinarity.
2. In terms of studying *practices*, modes and spaces of inter- and transdisciplinary collaboration, contributions should comprise a thick description (Geertz, 1973) of the problem spaces in different regions to examine what it means to do inter- and transdisciplinary research, how they are performed and where they take place.
3. In terms of contributing to current *policy* debates on the impacts of and funding for inter- and transdisciplinary research, new bridges need to be built for the funding and policy sectors by tracing intersections between practice and policy that lead to impactful inter- and transdisciplinary research.

This work leads to improved policies for inter- and transdisciplinary research and education grounded in a deep understanding of societal challenges (Vienni-Baptista et al., 2022). The cultural perspective I propose provides unique access to the dynamics of the messy, real-world environment of "doing" of inter- and transdisciplinarity, including the misunderstandings, aims and aspirations of those in academia and policy, to understand how they can better address societal challenges and provide firm foundations for scholarly understanding and the advancement of knowledge.[1]

Bianca Vienni-Baptista

Note

1. The work done by Bianca Vienni-Baptista is framed within the project "Investigating interdisciplinarity and transdisciplinarity: intersections of practices, culture(s) and

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